

Prince Sultan University College of Computer and Information Sciences			
Semester Project			
Course's Information:			
Course	CYS401: Fundamentals of Cybersecurity		
Semester	261		
Program	IS, CS, & SE		
Instructor	Dr. Yazan		
	Email: yalshboul@psu.edu.sa	Phone: 011-4948424	
Coursework Type	Group Project		
Percentage	15%		
Hand-out Date	Sep 13 th , 2026		
Due Date	Nov 4 th , 2026		
Students' Declaration:			
<i>We declare that:</i> 1. We understand what is meant by plagiarism 2. This assignment is all our own work and we have acknowledged any use of the published or unpublished works of other people. 3. We hold a copy of this assignment which we can produce if the original is lost or damaged			
Name	ID	Signature/Date	Marks
Learning Outcomes Assessed:			
- CLO 1: Understand the fundamentals of cyber security: policies, cryptographic systems, networking and communication, biometric systems, identity, authentication and authorizations management and techniques. - CLO 2: Analyze the security models and capabilities for specific security needs. - CLO 3: Describe the legal, ethical and professional the security principles and practices to the environmental, hardware, software, and human components of an information systems. - CLO 4: Apply appropriate tools to prevent, detect, react, and recover from attacks.			
Penalty for late submission:			
- 1 day – minus 20% of total mark awarded. - 2 days – minus 50% of total mark awarded. - 3 days – 0 mark for this piece of coursework.			

Criteria	Rubric Description			Given Marks
	Low Performance	Moderate Performance	Good Performance	
Depth of knowledge	-The ideas/concepts are not clear or not illustrated enough. (1-7)	-Most of the ideas/concepts are explained and supported with some brief examples. (8-14)	Exceptionally well-presented and argued; ideas/concepts are detailed, well-developed, supported with specific evidence & facts, as well as examples and specific details. (15-20)	
Presentation skills	Unable to explain or present; weak presentation skills (0-5)	Moderate explanation, acceptable presentation skills (6-11)	Good explanation and formal presentation (12-20)	
Analysis and opinions	Limited number of examples provided in the report and unjustified properly following relevant rules. Brief explanation presented. (0-5)	Variety of examples are provided in the report and justified properly following relevant rules. Some detail explanations are required. (6-12)	Good examples are provided in the report consist of good arguments and explanation. Justification is according to relevant rules. (13-20)	
Citations, references, overall structure, readability and completeness of the report	Improper citation, Incomplete paragraphs, and Poor structure of reports. (1-7)	Report is readable and structured. Most of references are cited properly. (8-14)	Report is readable and well-structured. The report covers all the aspects of discussion. All the references are cited properly. (15-20)	
Conclusion and recommendations	The conclusion lacks the main points discussed in the report, and has no recommendations (0-5)	The conclusion covers most of the main points discussed in the report, and has some recommendations (6-13)	The conclusion covers all the main points discussed in the report, and has strong recommendations (14-20)	
TOTAL MARKS (100)				

Projects Description

Important Instructions:

1. A proper interface should be implemented for demonstration.
2. Students need to follow the specified number of members per group.
3. Students are allowed to use the code from GitHub.
4. Students can choose any language for implementation for the projects.

1. Integrity checker (3-4 Students)

The Integrity Checker aims to provide security for operating systems. The tool will ensure the integrity and security of system files by verifying their integrity against known hashes or checksums. It will regularly scan the operating system files and compare them with a trusted database to detect any unauthorized modifications or tampering. The integrity checker will provide real-time alerts and notifications for any discrepancies found, enabling administrators or users to take immediate action.

FILE INTEGRITY MONITORING



2. Simple Malware Scanner using Yara (3-4 Students)

The project aims to create a simple malware scanner utilizing the Yara framework. The tool will scan files and directories using Yara rulesets containing known malware patterns and signatures. The scanner will compare the patterns in the rulesets with the content of the files, allowing for the detection of malicious files and potentially harmful software. It will provide users with real-time notifications and reports on the presence of any identified malware. The project aims to offer a user-friendly and efficient solution for detecting and mitigating malware threats using the power of Yara's pattern-matching capabilities.

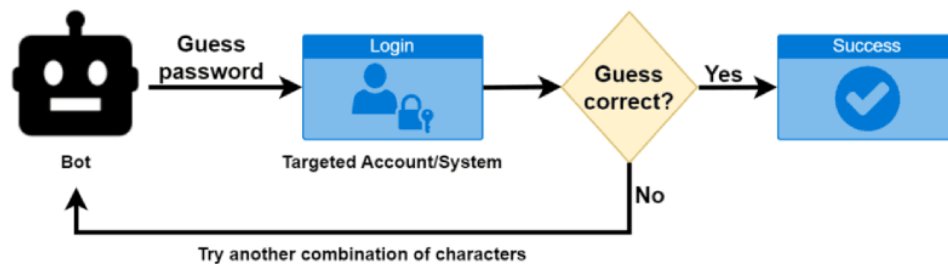
```

1 rule DarkRAT
2 {
3   meta:
4     date = "11/02/2015"
5     tags = "rat"
6   strings:
7     $a = "@190Gdark199Gcodex@"
8     $b = "SHEmptyRecycleBinA"
9     $c = "mciSendStringA"
10    $d = "add_Shutdown"
11    $e = "get_SaveMySettingsOnExit"
12    $f = "get_SpecialDirectories"
13   condition:
14     all of them
15 }
  
```

3. Simple non-hashed Password cracker (3-4 Students)

The simple non-hashed password cracker tool will focus on cracking passwords stored in plaintext or weakly encrypted formats. It will utilize various techniques such as brute force, dictionary attacks, and pattern matching to attempt to reveal the original passwords. The cracker will provide a user-friendly interface for inputting password files or encrypted data and will employ efficient algorithms to optimize the cracking process.

How Do Brute Force Attacks Work?



4. Keylogging (Spyware) (3-4 Students)

In computing, a keylogger is a piece of software that monitors all the keystrokes that take place on a computer's keyboard. It monitors previously pressed and currently active keys to record user activity. You can consider the following project on the concept of keylogging:

- Create a script that can record keystrokes.
- Create and bind a keylogger with legitimate application/software to record user activity, basically spyware.
- Create a script which can detect the presence of a keylogger on endpoint. (Optional)

5. Implementing a Random Password Generator (3-4 Students)

A random password generator is a software program, hardware device, or online tool that automatically generates a password using parameters that a user sets, including mixed-case letters, numbers, symbols, pronounceability, length, and strength.

You need to create/ implement the Random Number Generator (C++, Java, Python), that asks you for the length of the password and generate a secure random password for you. Sample Random Passwords: **BP#/P-pQ8p2_, Z9tOh|ES*GOX, 1b-‘\$W^\$)e/I, q8uQPjgyzp%:, [“cB6}%gvl~a.**

After generating the random password, you must check the password strength of that password using some online brute force calculator. The sample interface is given below:



Password Length: (6-64 Chars)

Include Small Letter: ☒ (e.g:abcd)

Include Capital Letter: ☒ (e.g:ABCD)

Include Numbers: ☒ (e.g:01393)

Include Punctuation: ☒ (e.g: !@#\$\$%^)

Quantity: (1-100 Passwords)

6. Access user Webcam, Microphone & Mobile Location (3-4 Students)

Access user Webcam, Microphone & get Mobile Location using a Link with Storm- Breaker. You can easily do this project in Kali Linux. All you need to do is clone the Storm Breaker from <https://github.com/ultrasecurity/Storm-Breaker> into Kali Linux (terminal).

After completing the project, you need to write a list of security controls/ recommendations to keep yourself secure from illegitimate access through webcam, microphone, or mobile location. Try to implement any one security controls against this attack.

7. Generate Captcha Code Using Java and Integrate with Spring Boot (3-4 Students)

CAPTCHA technology authenticates that a real person is accessing web content to block spammers and bots that try to automatically harvest email addresses or try to automatically sign up for access to websites, blogs or forums. CAPTCHA blocks automated systems, which can't read the distorted letters in the graphics.

Generate captcha is very simple, it basically validates two values one which we kept in session when we are rendering the captcha and one which user enters, so before going further after login we validate captcha values like session captcha value and user entered value so if both are equals then we are allowing user to go further, and if does not match then return back with proper message like captcha is invalid please try again.

Here are the steps which you need to follow to generate our own captcha. You need a servlet and which we directly call from login page and that will return an image, so image creation done by java awt.

8. Graphical Password Authentication (3-4 Students)

Passwords are ubiquitous today on any platform, on possibly any website. But to remember so difficult passwords and that too on numerous websites seems daunting and therefore you can devise a project illustrating graphical password strategy. In this method, the user is required to select some images (let's say different chocolates) in a specific pattern (for example dairy milk is followed by 5 stars which is in turn followed by KitKat and so on). Next time the user tries to log in, the images would have been shuffled, but the user will be required to follow the same pattern which was used initially. Every time the user will have to use the same

sequence while the images are placed in different ways. This type of authentication is difficult to break since neither brute force nor dictionary attacks could breach it.



9. Text Steganography Project (3-4 Students)

Steganography embeds a secret message inside an innocent looking cover medium, stealthily, without creating any attention. The cover medium used can be a text, image, audio, video, network packets, etc. In this desktop application there are 3 interfaces: Registration, Login, encode & decode. User needs to register first and he/she can login to the application while the user supposed to enter a PIN as a multifactor authentication when he/she login's with username and password. After logging in, user can write text and encode it and can share with the person whom he/she wants to send the secret message. This is how the text steganography, hide text within text works.

10. Image Encryption Using AES Algorithm (3-4 Students)

Today almost all digital services like internet communication, medical and military imaging systems, multimedia system requires reliable security in storage and transmission of digital images. Due to faster growth in multimedia technology, internet and cell phones, there is a need for security in digital images. Therefore, there is a need for image encryption techniques in order to hide images from such attacks. In this system we use AES (Advanced Encryption Technique) in order to hide image. Such Encryption technique helps to avoid intrusion attacks.

11. Frequency Analysis Attack using Python (3-4 Students)

The frequency analysis process consists of the following steps:

- Create a list of plaintext frequencies, sorted by frequency.
- Create a list of encrypted text frequencies, sorted by frequency.
- Merge these lists to create an estimated mapping between encrypted letters and plaintext letters.
- Use this mapping to create an approximate decryption of the encrypted text.

12. Password Manager (3-4 Students)

Develop a secure password manager application that stores and generates strong, unique passwords for various accounts. Implement encryption techniques to safeguard the passwords and encourage users to practice better password hygiene. This project aims to reduce the risk of password breaches and enhance overall account security.

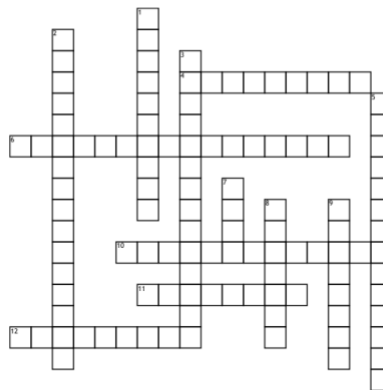
13. Cybersecurity Quiz Application (3-4 Students)

Build an engaging quiz app with questions about cybersecurity concepts, best practices, and common threats. The app should help users test their knowledge and raise awareness about essential cybersecurity principles.

14. Cybersecurity Crossword Puzzle (3-4 Students)

Design a crossword puzzle with cybersecurity-related clues to reinforce your knowledge of key terms and concepts.

Cybersecurity Crossword Puzzle



Across

4. What is one thing cyberattacks aim at?
6. What is on career that deals with cybersecurity?
10. What is computer security?
11. What is one thing cybersecurities protect?

Down

1. What is one benefit of cybersecurity?
2. What is one thing cybersecurity is related to?
3. What is one cybersecurity threat vector?
12. What is one thing related to cybersecurity?

Across

5. What is one thing cybersecurity can prevent?
7. What is one example of network security?
8. What type of challenges do cybersecurities face?
9. What is one type of cybersecurity threat?

Word Bank

cybersecurity	Accessing	Breaches	VPNs
Data Theft	Physical Security	Identity Thefts	Phishing
Malvertisement	Protection	Hackers	Security Engineer

15. Website Scraper (3-4 Students)

A Web Scraper is a program that scrapes or collects data from websites with absolute accuracy. Let us imagine that we are developing a web scraper that will go to Twitter and collect the content of tweets. In its simplest form, web scraping is the act of collecting data from the internet in any form. Web scraping, on the other hand, enables you to gather information in huge volumes by using bots on a large scale. Crawlers or spiders are bots that scan the source code of a particular web page and tag data based on some predefined parameters. Following that, the data extractor gathers the enclosed data and exports it to a spreadsheet file.

16. Brute-force password cracker (3-4 Students)

Create a programme that uses the brute force attack in order to crack a password. Helps in understanding how important it is to choose a strong password which is difficult to crack.

17. Digital Footprint Visualizer (3-4 Students)

Nowadays, people share so much of themselves online that they forget how much unnecessary information they are giving to the internet. This project will help you understand your online presence by creating a simple tool that shows where your data is appearing across different websites and social media platforms. This tool collects public information and then displays it visually so that one can see one's exposure. This project will definitely help society become more aware of digital privacy while supporting individuals in controlling their online identities

18. Social Engineering Prevention Toolkit (3-4 Students)

Social engineering is one of the quickest ways that hackers use to fool people into providing their private data. This project aims to build a social engineering prevention toolkit, which is an excellent university project that requires a combination of your technical and social skills. This will help you recognise suspicious messages, fake emails, or misleading links. You can also add short quizzes, videos and an alert system to educate the audience. This will promote a smarter and safer online behaviour for everyone, especially for the newbies.

19. Password Strength Evolution Tracker (3-4 Students)

Weak passwords are still a major reason behind many cyberattacks. This project lets users track how their password habits improve over time. This tool will guide you with the actual feedback of your password, like its strength, and give potential suggestions for improvement, if required. This outcome promotes a strong password culture and motivates people to manage their accounts wisely.

20. Cyber Hygiene Education Game (3-4 Students)

With cybercrime on the rise, there is a clear need for creative educational tools. You can make learning about cybersecurity doesn't have to be boring with this tool. Through this software or app, you can create a simple game for teaching online safety through fun challenges. This idea is great for schools and beginners who want to learn while playing. It will work because it turns security learning into an enjoyable experience.

21. Children's Online Safety Analyser (3-4 Students)

Children are spending more time online than ever before, but not all websites are safe for them. This project aims to build a web tool that scans websites or apps and checks if they collect unnecessary data or show harmful content. Please keep it simple, with clear ratings and warnings, blocking inappropriate search results based on the keywords or categories. It protects children's privacy and builds a safer online world for the next generation while teaching students the value of ethical technology use.

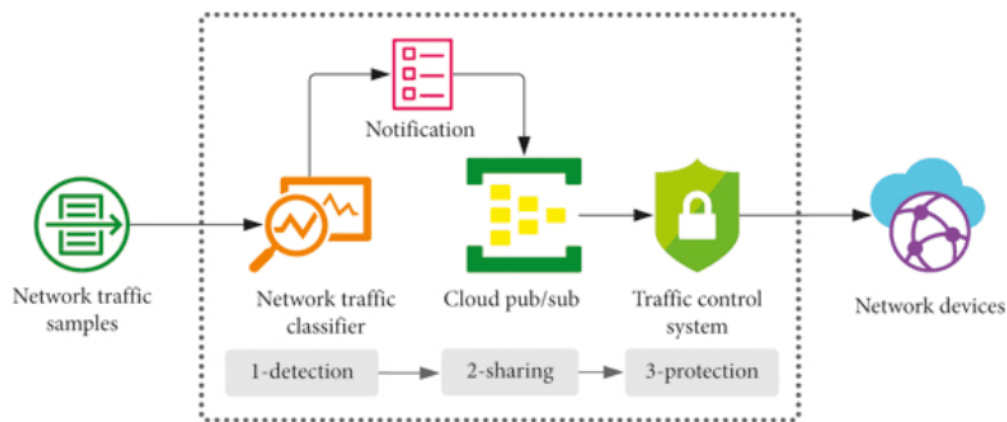
22. Incident Response Automation Bot (3-4 Students)

To make this project, you first need to define a use case, such as a phishing attack or malware infection. Then, choose the programming language; Python is the most recommended one. Select a communication platform like

Slack or Discord to send instant alerts. This protects businesses, reduces the financial cost of breaches, and safeguards the personal data of customers and employees.

23. DOS detection (4 Students)

The DoS (Denial of Service) attack detection tool will be specifically designed for beginners. This tool will provide an intuitive interface and simple configuration options to assist users in detecting and mitigating DoS attacks. It will monitor network traffic patterns, analyze abnormal traffic spikes, and identify potential signs of a DoS attack. The tool will generate real-time alerts and notifications, allowing users to take immediate action to mitigate the attack. The project's objective is to empower beginner-level users with an accessible and effective solution for detecting and responding to DoS attacks, enhancing their systems' overall security and availability.



[Source code](#)

24. Phishing Email Detection Tool (3–4 Students):

Develop a simple tool that analyzes emails for phishing indicators such as suspicious URLs, sender spoofing, keywords, and header anomalies. The system should flag potential phishing emails and provide explanations to educate users on phishing techniques.

25. File Encryption & Secure File Sharing Tool (3–4 Students)

Create a desktop or web-based application that allows users to encrypt files before sharing them and decrypt them securely on the receiving end. Implement symmetric encryption (e.g., AES) and basic key management features.

26. Vulnerability Scanner for Web Applications (3–4 Students)

Develop a lightweight scanner that checks for common web vulnerabilities such as weak passwords, open ports, or missing HTTPS headers. The tool should generate a simple security report with recommendations.

27. Interactive Cybersecurity Awareness Assistant (3–4 Students)

This project involves developing an interactive awareness tool that educates users about common cybersecurity threats and safe practices through real-life scenarios. The system can present users with short situations such as phishing emails, weak password choices, unsafe Wi-Fi usage, or social engineering attempts, and ask them to make decisions. Based on the user's choices, the tool provides immediate feedback, explains the risks involved, and recommends best security practices.

The application should include:

- Short, scenario-based questions and simulations
- Explanations of correct vs. incorrect decisions
- A simple scoring or progress-tracking mechanism
- A user-friendly interface for demonstration

This project helps users recognize threats, understand consequences, and build safer digital behavior, making it an effective tool for improving cybersecurity awareness and supporting ethical and professional responsibility in computing practice.

28. Mini Capture-The-Flag (CTF) Security Lab Platform (3–4 Students)

This project aims to design and implement a mini Capture-The-Flag (CTF) security lab platform that simulates real-world cybersecurity challenges commonly found in CTF competitions. The platform will provide users with a set of guided security challenges covering fundamental cybersecurity topics such as authentication weaknesses, insecure configurations, phishing awareness, cryptography basics, and ethical decision-making.

Each challenge will represent a realistic security scenario where users must analyze the situation, identify the security issue, and submit a correct “flag” to demonstrate understanding. The system will automatically validate flags, track user progress, and display feedback explaining the correct security concepts and best practices.

The platform should include:

- A user-friendly web or desktop interface to navigate challenges
- Multiple challenge categories (e.g., Web, Crypto, Awareness, Ethics)



- Flag validation and progress tracking
- Hints or explanations to support learning
- A simple scoreboard or completion status

This project emphasizes hands-on learning, critical thinking, and security awareness, while introducing students to CTF-style problem solving in a controlled, ethical, and educational environment aligned with course learning outcomes.

Project Specification

You must strictly follow the following specifications.

1. This project is worth **15%** of the coursework marks (*Division: 5 Marks for Report, 5 Marks for demonstration/ presentation and 5 marks for individual assessment during presentation*).
2. The requirements of the project Report (**Read Rubrics**):
 - **Phase-I:** Introduction of the topic in relation to security. To assess common security risks, threats, attacks and vulnerabilities in an identified assets/ area of an organization.
 - **Phase-II:** Literature Review: Existing/ Previous/ Substitute methods of implementing the solution. Any security incidents happened in that area.
 - **Phase-III:** Implementation + requirements (Hardware & Software). Interface screenshots with proper description. Include code as well at the end of the report in Appendix.
 - **Phase-IV:** To apply all that you have learned in this class and put it into practice by writing a document. To write a good documentation using proper document formatting, and proper referencing (**End of the Document: Application & Recommendations to overcome the threats**).
 - **Phase V:** Students are required to present & demonstrate their project within a 10-minutes PowerPoint presentation including questions and answer session.
3. Below are guidelines for the format of the documentation:
 - The Project should be between **13 ~ 15** pages (strictly).
 - Submit your project report & presentation on MOODLE before deadline
 - Headings should be capitalized, boldface, and left aligned.
 - Line spacing should be justified with single space.
 - Document texts must be in Times New Roman size 12.
 - Include a page number on each page.
 - Spell check and proofread you work.



4. Below are guidelines for writing the document:

- Document Cover Page must include: University Logo, Course Code, Course Title, Project Title, Group member names, ID and Instructor Name.
- A content page (contains titles, sub-titles, and page number reference)
- Introduction (a paragraph to give an overview of the project)
- Body part of the documentation (write your comparative analysis and user policies) (if applicable)
- Conclusion (a paragraph of summary of the project)
- Reference page (all references must be referred to in the reference page)

END!!